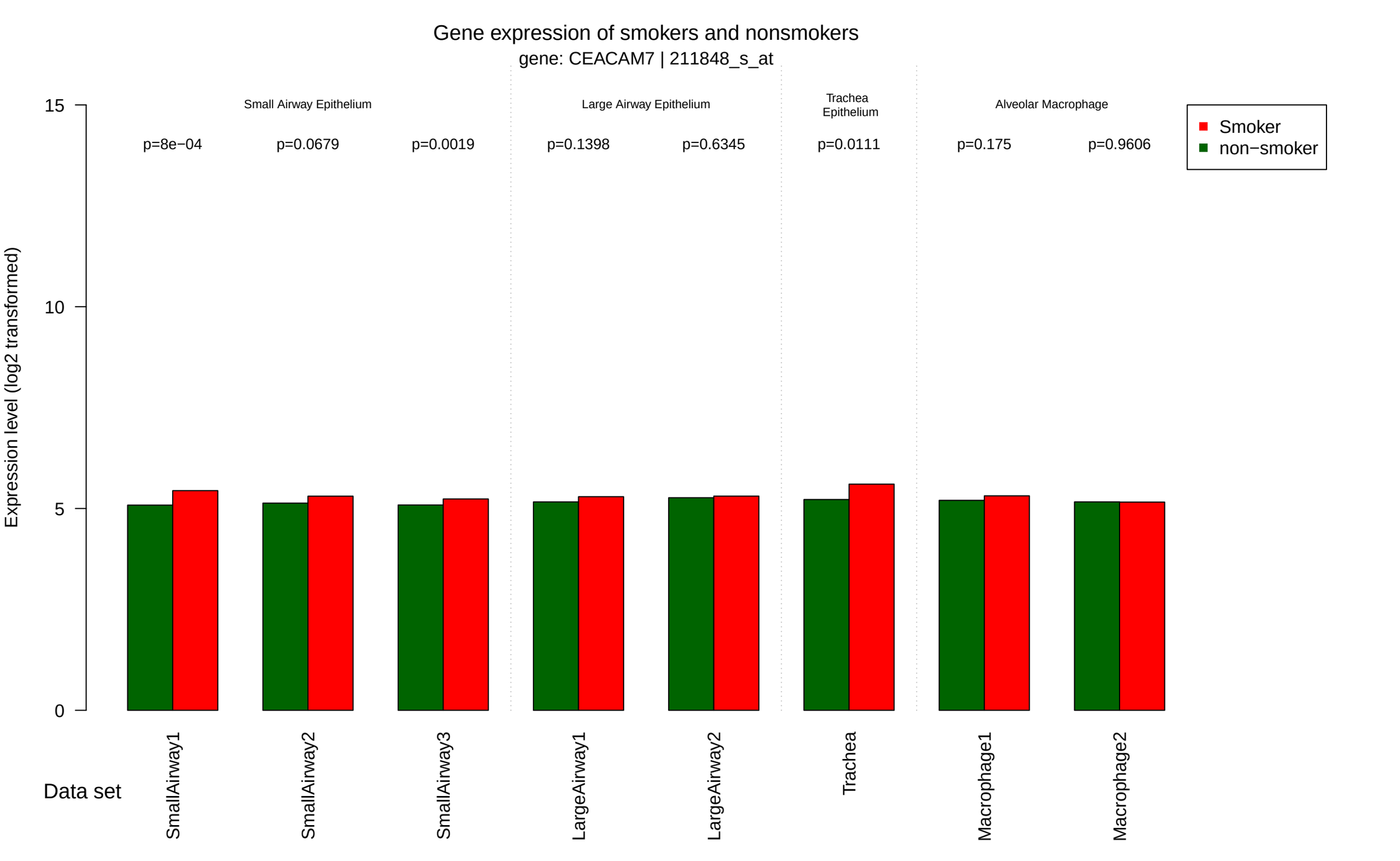
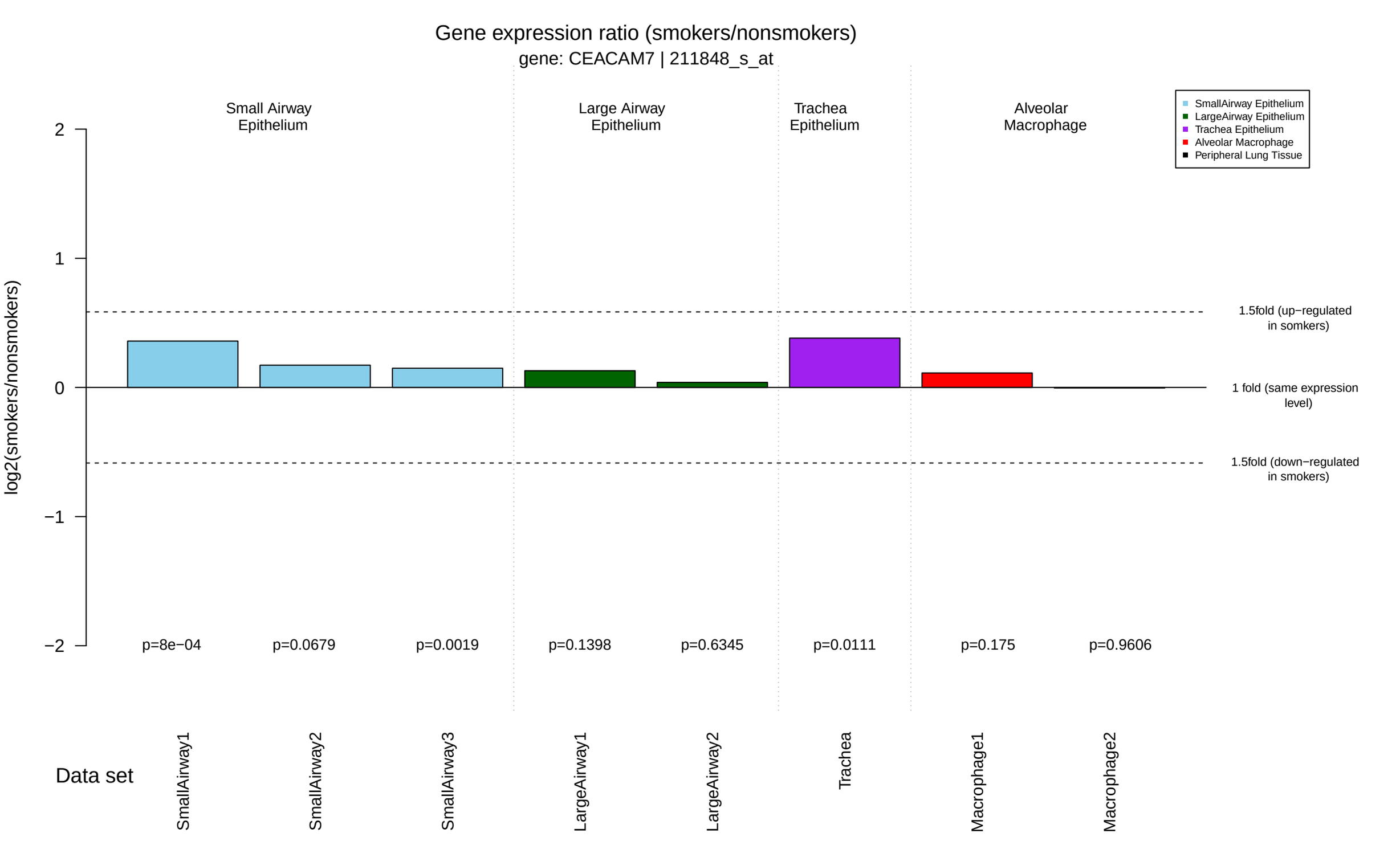
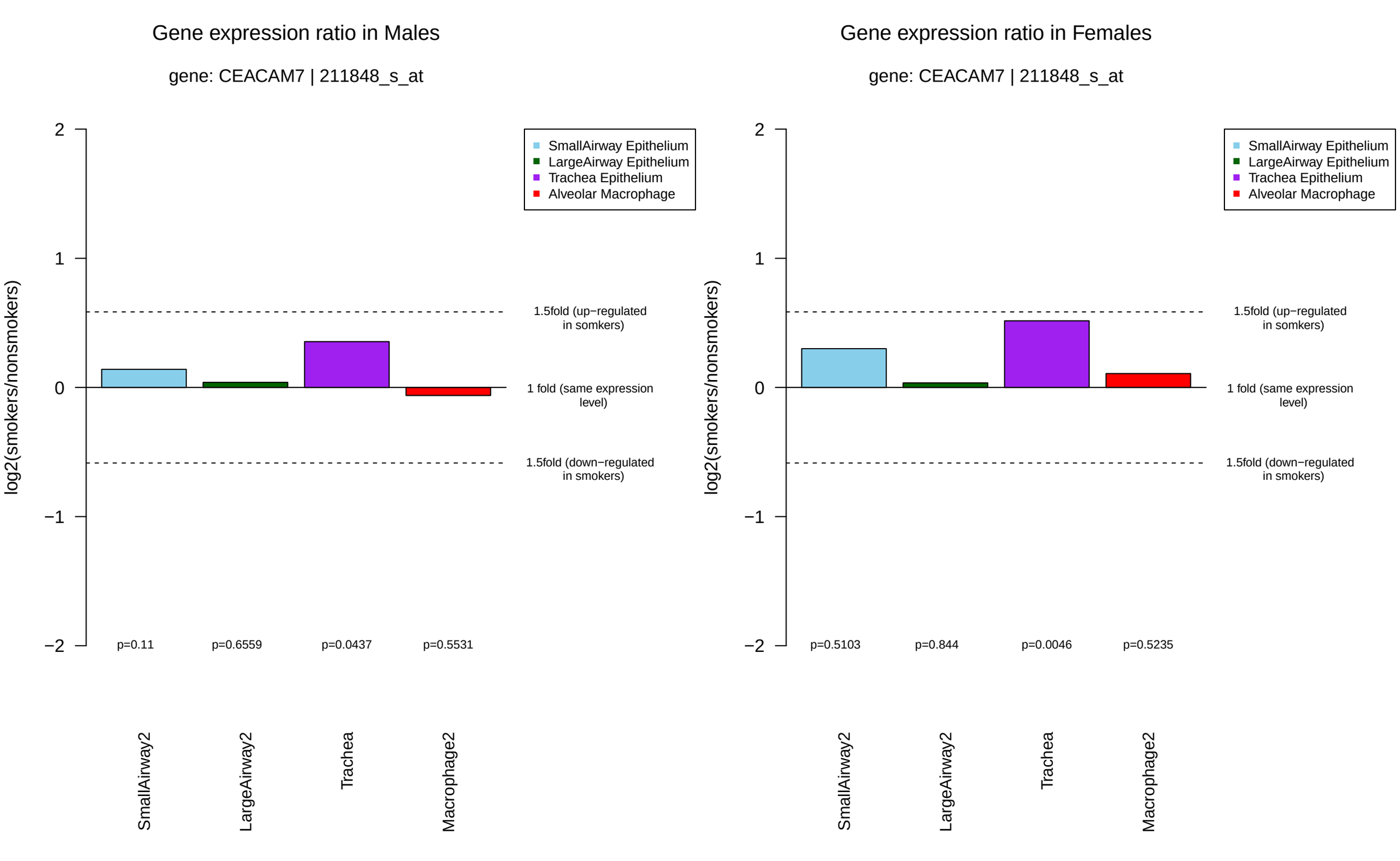
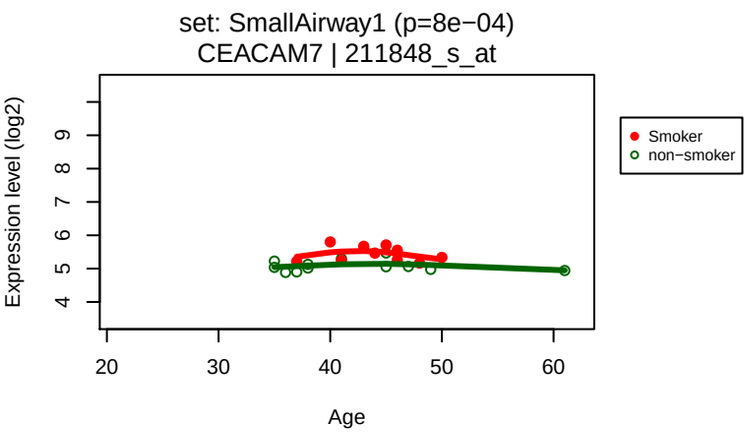
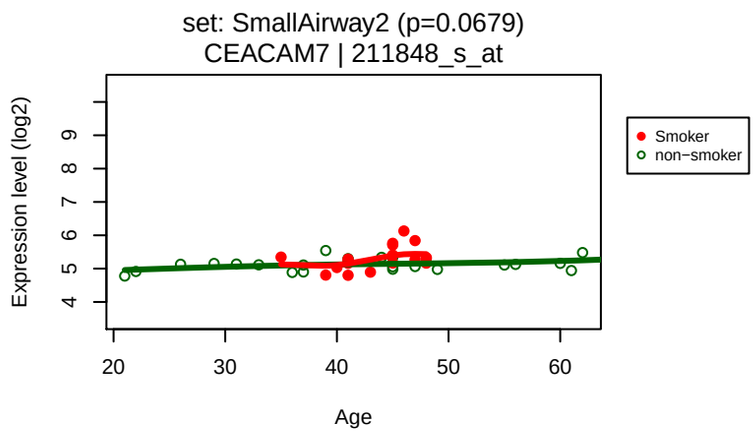
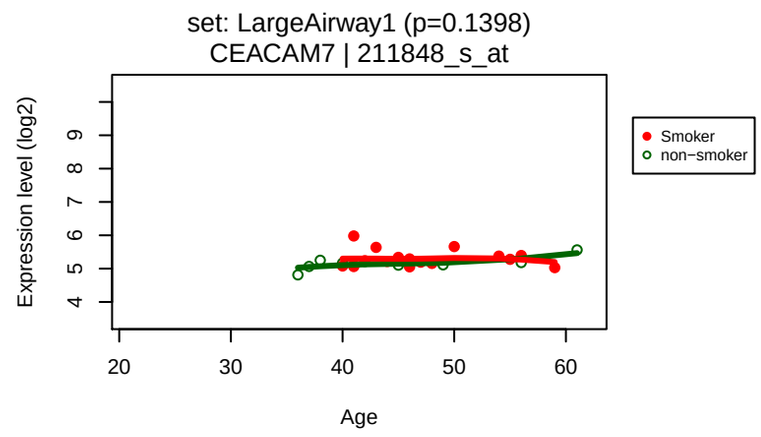

Gene expression of smokers and nonsmokers
gene: CEACAM7 | 211848_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=8e−04
p=0.0679
p=0.0019
p=0.1398
p=0.6345
p=0.0111
p=0.175
p=0.9606
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

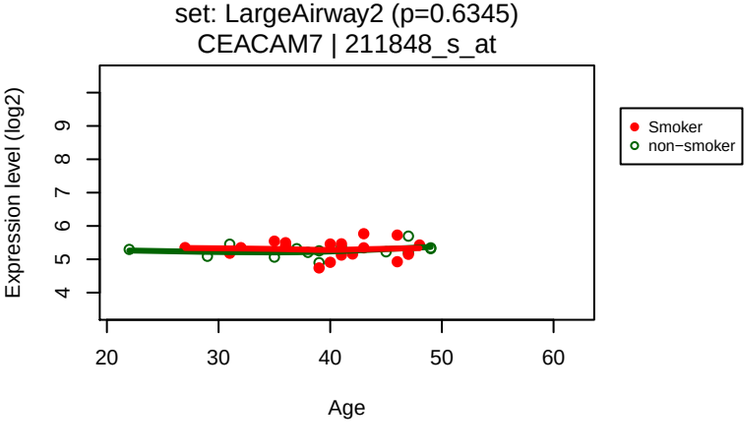

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM7 | 211848_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=8e−04
p=0.0679
p=0.0019
p=0.1398
p=0.6345
p=0.0111
p=0.175
p=0.9606
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

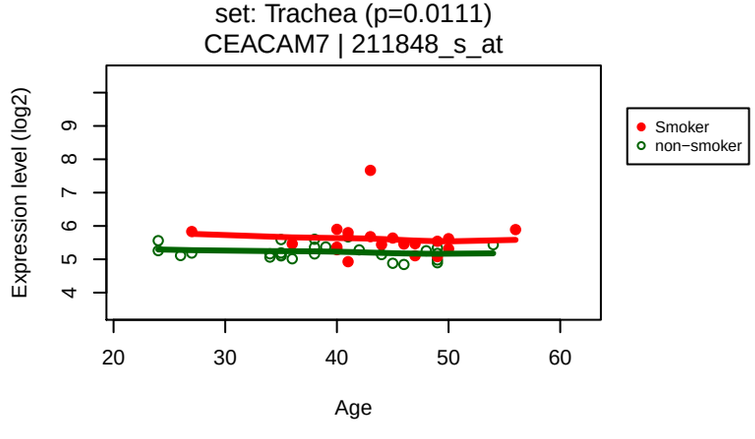

Gene expression ratio in Males
gene: CEACAM7 | 211848_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.11
p=0.6559
p=0.0437
p=0.5531
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM7 | 211848_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5103
p=0.844
p=0.0046
p=0.5235
SmallAirway2
LargeAirway2
Trachea
Macrophage2

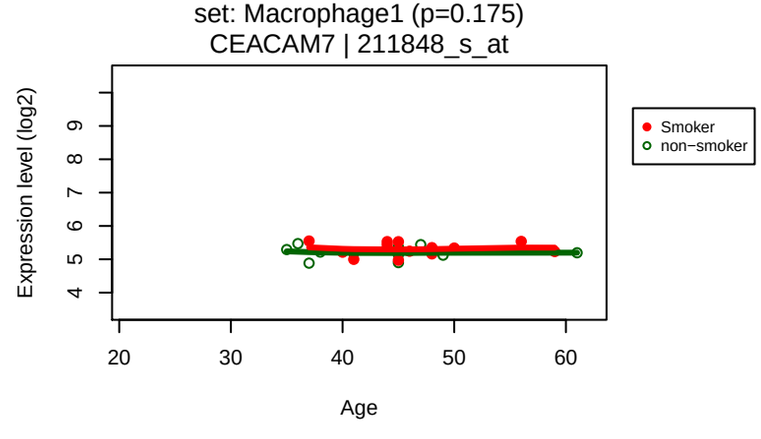

set: SmallAirway1 (p=8e−04)
CEACAM7 | 211848_s_at
Expression level (log2)
Age
Smoker
non−smoker

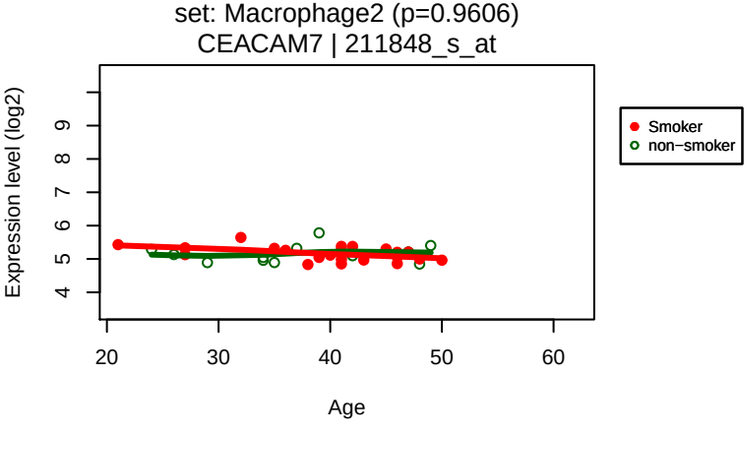

set: SmallAirway2 (p=0.0679)
CEACAM7 | 211848_s_at
Expression level (log2)
Age
Smoker
non−smoker

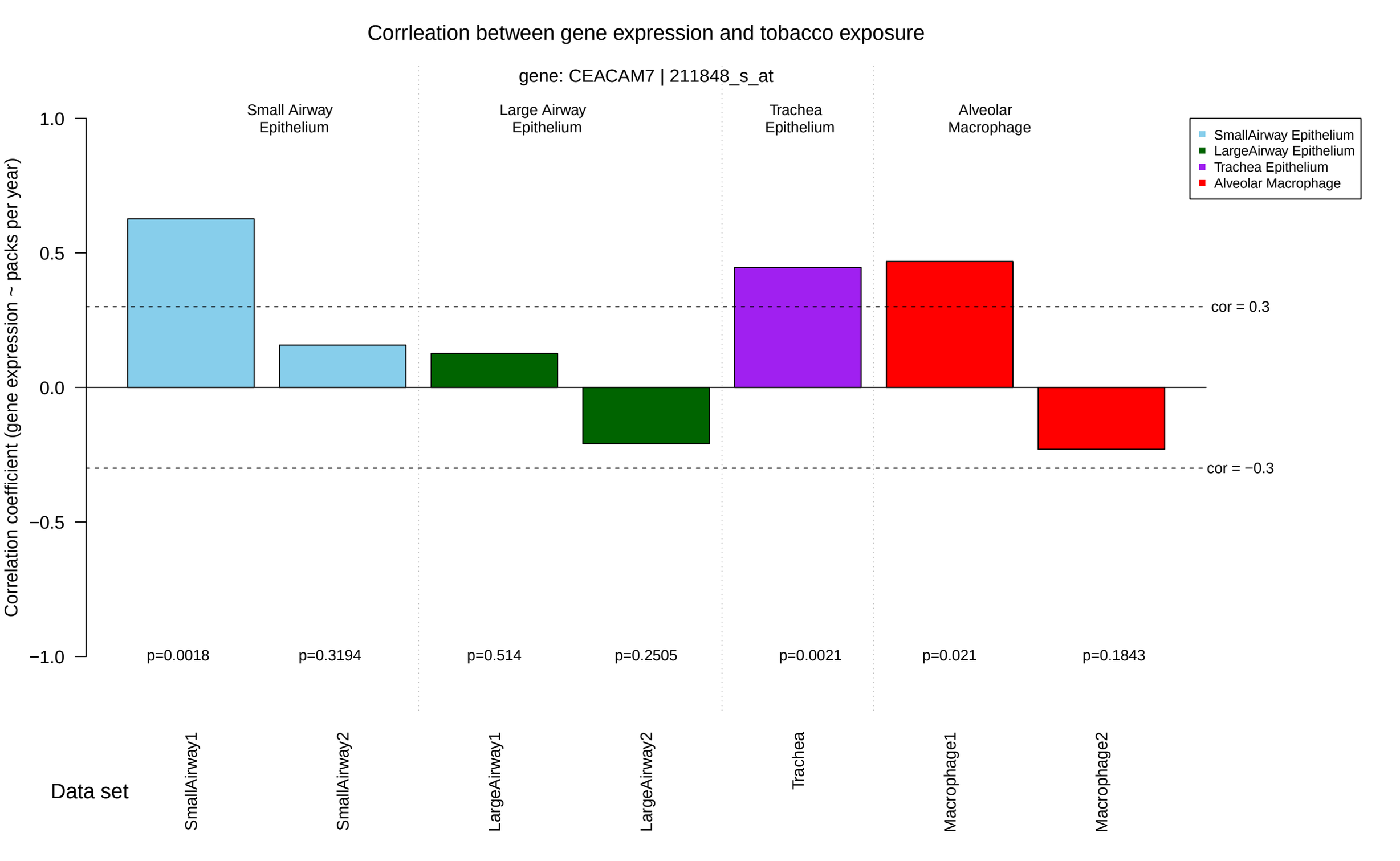

set: LargeAirway1 (p=0.1398)
CEACAM7 | 211848_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6345)
CEACAM7 | 211848_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0111)
CEACAM7 | 211848_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.175)
CEACAM7 | 211848_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9606)
CEACAM7 | 211848_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM7 | 211848_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0018
p=0.3194
p=0.514
p=0.2505
p=0.0021
p=0.021
p=0.1843
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2